Probability and Stochastic Processes

Juan YANG

School of Sciences, BUPT

June 2025

Chapter 8 Other Stochastic Processes

Independent increment process is a special kind of Markov process. Poisson processes and Wiener processes are the two most important independent increment processes and they are the basis of Johnson noise theory.

- Independent-Increment Processes
- Poisson Process
- Gaussian Processes
- Brownian Motion and Wiener Process

Contents

1. Independent-Increment Processes
2. Poisson Process
3. Gaussian Processes
4. Brownian Motion and Wiener Processes

Independent-Increment Processes

Definition

A stochastic process $\{X_t\}_{t\geqslant 0}$ *is said to have independent increments if the set of the following random variables,*

$$X_{s_1} - X_0,\ X_{t_1} - X_{s_1},\ X_{t_2} - X_{s_2},\ \cdots,\ X_{t_n} - X_{s_n}$$

are independent for all $0 \leqslant s_1 < t_1 \leqslant s_2 < t_2 \leqslant \cdots \leqslant s_n < t_n < +\infty$, $n \geqslant 1$. *Furthermore, if the distribution function of* $X_{t+s} - X_s$ *is independent of* s, *we say that* $\{X_t\}_{t\geqslant 0}$ *has stationary independent increments. The stochastic process* $\{X_t\}_{t\geqslant 0}$ *is called stationary independent-increment process or homogeneous independent-increment process.*

Independent-Increment Processes

Theorem

Suppose that $X_0 = 0$. *For the independent-increment process* $\{X_t\}_{t \geqslant 0}$, *its finite-dimensional distribution function can be determined by the distribution function of* $X_t - X_s$, $0 \leqslant s < t$.

Theorem

Suppose that $\{X_t\}_{t \geqslant 0}$ *is an independent-increment process and* $X_0 = 0$. *Then the covariance function* $C_X(s,t) = Var_X[\min(s,t)]$.

Contents

1. Independent-Increment Processes
2. Poisson Process
3. Gaussian Processes
4. Brownian Motion and Wiener Processes

Poisson Processes

A Poisson process is a stochastic process which counts the number of events, which occur in a given time interval. The process is named after the French mathematician Siméon Denis Poisson. Examples to keep in mind are the moments at which telephone calls are received in a call centre, the times of customer/job arrivals at queues, or the moments at which Peking is hit by an earthquake. It is one of the most important models used among many phenomena.

Poisson Processes

When we try to model the occurrences of the various processes described above, there are a number of characteristics that we may want to build in. To name a few:

(i) There is a certain amount of regularity in the processes described above. Although individual earthquakes are impossible to predict and certainly do not occur in a strictly regular pattern, there is perhaps some statistical regularity in the sense that when we observe the earthquakes during 10 years, say, without knowing the absolute time frame, then we have no way to decide whether we observe the time period 1910-1920 or 1990-2000. In probabilistic terms, the process is stationary in time. In other words, the course of time should not change the probabilistic properties of the process. That means we suppose the process is stationary.

Poisson Processes

(ii) The fact that there is an occurrence at a particular time, says nothing about the probability of an occurrence at, or around, a later or earlier time. In other words, there seems to be some kind of independence with respect to various occurrences.

(iii) The next occurrence can not be predicted from current and past information. In other words, the process of occurrences seems to have no memory. The fact that something happened in the past has no effect on the probabilities for future occurrences.

(iv) In each finite time interval, there are only finitely many occurrences.

Poisson Processes

Keep these characteristics in mind and we will build a mathematical model for this kind of processes by different approaches. We think that it is very instructive to travel them all.

Approach 1. (via waiting times) Perhaps the most natural thing to do is to concentrate on the waiting times.
Point (i) above suggests that the waiting time distribution should be the same at all times. For example, the waiting time between the 6th and 7th should have the same distribution as the waiting time between the 12th and 13th.

Poisson Processes

Point (iii) above suggests that the process should have no memory. We have come across a continuous distribution with a certain lack of memory property. Indeed, we know that the exponential distribution has no memory in the sense that when X has such an exponential distribution, then

$$P(X > s + t | X > t) = P(X > s).$$

If we think of X as the waiting time between successive occurrences, then this formula expresses the idea that we have waited already t time units does not change the probability that we have to wait another s time units. This property makes the exponential distribution a serious candidate for the waiting times between successive occurrences.

Poisson Processes

Consider independent random variables $X_1, X_2, \cdots$, which are exponentially distributed with parameter λ, that is, they have p.d.f. f given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{for } x \leqslant 0. \end{cases}$$

The first occurrence is at time X_1, the second at time $X_1 + X_2$, etc. The sequence $\{X_n, n = 1, 2, \cdots\}$ is called the *sequence of interarrival times.* For instance, if $X_1 = 5$ and $X_2 = 12$, then the first event of the Poisson process would have occurred at time 5 and the second at time 12. In general, we can define

$$S_n = \sum_{i=1}^{n} X_i,$$

and then S_n should be thought of as the moment of the nth occurrence.

Poisson Processes

In principle, this defines the model completely. For now, it is perhaps good to look at the parameter λ. Since $E(X_i) = \lambda$, we see that a high λ corresponds to a small average waiting time. Therefore, it makes sense to call λ the intensity of the process.

Approach 2. (via the number of occurrences)

Approach 3. (via differential equations)

Poisson Processes

Definition

Let $X_1, X_2, \cdots$ *be independent and identically distributed random variables with an exponential distribution with parameter* $\lambda > 0$. *Let* $S_0 = 0$ *and*

$$S_n = \sum_{i=1}^{n} X_i \quad \text{for } n = 1, 2, \cdots ,$$

We call X_i *the* i*th interarrival time, and* S_n *the moment of the* n*th occurrence.*
For $t \geqslant 0$, *let* $N_t = \max\{n : S_n \leqslant t\}$ *and* $N_0 = 0$. *We call* N_t *a Poisson process with intensity* λ.

Poisson Processes

Definition

The counting process $\{N_t\}_{t \geqslant 0}$ *is said to be a Poisson process having rate* λ, $\lambda > 0$, *if*
(i) $N_0 = 0$,
(ii) *the process has independent increments,*
(iii) *the number of events in any interval of length* t *is Poisson distributed with mean* λt. *That is, for all* $s, t \geqslant 0$,

$$P\{N_{t+s} - N_s = n\} = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \ n = 0, 1, \cdots$$

Poisson Processes

Note that it follows from condition (iii) that a Poisson process has stationary increments.

Property

Let N_t be a Poisson process with rate λ. Then
(i) $\mu_N(t) = \lambda t$,
(ii) $C_N(s,t) = Var_N[\min(s,t)] = \lambda \min(s,t)$,
(iii) $R_N(s,t) = \lambda \min(s,t) + \lambda^2 st$,
where

$$\min(s,t) = \frac{s+t-|s-t|}{2}.$$

Poisson Processes

The time between t and S_{N_t+1} is not an ordinary waiting time; it is the remainder of the current waiting time at time t. You might expect that this remaining waiting time is in some sense "smaller" than an "ordinary" waiting time. However, the following surprising and fundamental result tells us that this is not the case. It tells us that the remaining waiting time at time t, and all subsequent waiting times are independent and have the same exponential distribution.

Theorem

Let $V_t := S_{N_t+1} - t$, and let for all $i = 2, 3, \cdots$, $Z_i := X_{N_t+i}$. Then for all m, the random variables $V_t, Z_2, \cdots, Z_m$ are independent and identically distributed, with an exponential distribution with parameter λ.

Poisson Processes

Example

Let N_t be a Poisson process with parameter λ. Find the values of $E(N_t^2)$ and $E[(N_t - N_s)^2]$, for $t > s$.

Example

Defects occur along an undersea cable according to a Poisson process of rate $\lambda = 0.1$ per mile.
(a) What is the probability that no defects appear in the first two miles of cable?
(b) Given that there are no defects in the first two miles of cable, what is the conditional probability of no defects between mile points two and three?

Poisson Processes

Example

Customers arrive in a certain store according to a Poisson process of rate $\lambda = 4$ per hour. Given that the store opens at 9:00 A.M., what is the probability that exactly one customer has arrived by 9:30 and a total of five have arrived by 11:30 A.M.?

Example

Patients arrive at the doctor's office according to a Poisson process with rate $\lambda = 1/10$ minute. The doctor will not see a patient until at least three patients are in the waiting room. What is the probability that nobody is admitted to see the doctor in the first hour?

Contents

1. Independent-Increment Processes
2. Poisson Process
3. Gaussian Processes
4. Brownian Motion and Wiener Processes

Gaussian Processes

The fundamental characterization of a Gaussian process is that all the finite dimensional distributions have a multivariate normal (or Gaussian) distribution. It is completely determined by its mean and covariance functions. This property facilitates model fitting as only the first- and second-order moments of the process require specification. Gaussian processes can be used as building blocks to construct more complex models that are appropriate for non-Gaussian data.

Gaussian Processes

Definition

A real-valued stochastic process $\{X_t\}_{t \in T}$ *is said to be a Gaussian process if all the finite-dimensional distributions have a multivariate normal distribution. That is, for any choice of distinct values* $t_1, \cdots, t_k \in T$*, the random vector* $\boldsymbol{X} = (X_{t_1}, \cdots, X_{t_k})'$ *has a multivariate normal distribution with mean vector* $\mu = E(\boldsymbol{X})$ *and covariance matrix* $\Sigma = Cov(\boldsymbol{X}, \boldsymbol{X})$*, which will be denoted by*

$$\boldsymbol{X} \sim N(\mu, \Sigma).$$

Gaussian Processes

Provided the covariance matrix Σ is nonsingular, the random vector $\boldsymbol{X}$ has a Gaussian probability density function given by

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = (2\pi)^{-n/2}(det\Sigma)^{-1/2} \times \exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)'\Sigma^{-1}(\boldsymbol{x}-\mu)\right).$$

The mean and covariance functions of a Gaussian process are defined by

$$\mu_X(t) = E(X_t) \quad \text{and} \quad C_X(s,t) = Cov(X_s, X_t)$$

respectively.

Gaussian Processes

While Gaussian processes depend only on these two quantities, modeling can be difficult without introducing further simplifications on the form of the mean and covariance functions. The assumption of stationarity frequently provides the proper level of simplification without sacrificing much generalization. Moreover, after applying elementary transformations to the data, the assumption of stationarity of the transformed data is often quite plausible.

Gaussian Processes

A Gaussian time series $\{X_t\}$ is said to be stationary if

1. $\mu_X(t) = E(X_t) = \mu_X$ is independent of t, and

2. $C_X(t, t+h) = Cov(X_t, X_{t+h}) = C_X(h)$ is independent of t for all h.

Gaussian Processes

Property

For a wide-stationary Gaussian process $\{X_t\}$*, we have*
(i) $X_t \sim N(\mu_X, C_X(0))$ *for all* t*, and*
(ii) $(X_t, X_{t+h})'$ *has a bivariate normal distribution with covariance matrix*

$$\begin{pmatrix} C_X(0) & C_X(h) \\ C_X(h) & C_X(0) \end{pmatrix}$$

for all t *and* h*.*

Gaussian Processes

Property

(i) *Suppose that* $\{X_t\}_{t\in T}$ *is a Gaussian process. Then it is strict-sense stationary if and only if it is wide-sense stationary.*
(ii) $\{X_t\}_{t\in T}$ *is a Gaussian process if and only if any linear combination of any finite number of random variables has a normal distribution, i.e., for any* $t_1, t_2, \cdots, t_n \in T$ *and* $c_1, c_2, \cdots, c_n \in \mathbb{R}$, *the random variable* $c_1X_{t_1} + c_2X_{t_2} + \cdots + c_nX_{t_n}$ *has a (univariate) normal distribution.*
*(iii) *Suppose that* $\{X_t\}_{t\in T}$ *is a Gaussian process and* S_t *is a deterministic function. Then for all* t, $X_t + S_t$ *has a normal distribution.*

Gaussian Processes

Property

*(iv) *Suppose that a Gaussian process* $\{X_t\}_{t\in T}$ *is mean-square differentiable. Then its derivative* $\{X_t^{'}\}_{t\in T}$ *is also a Gaussian process.*

*(v) *Suppose that a Gaussian process* $\{X_t\}_{t\in T}$ *is mean-square integrable, then for all* a, b *and* $t \in T$,

$$Y_t = \int_a^t X_s ds, \quad \textit{and} \quad Z_t = \int_a^b X_s h(s,t) ds$$

are both Gaussian processes.

Gaussian Processes

Example

Suppose that $\{X_t\}_{t \in T}$ is a Gaussian process with the mean function $E(X_t) = 0$ and autocorrelation function $R_X(\tau) = \frac{1}{4}e^{-2|\tau|}$. For all t, find $P\{0.5 \leqslant X_t \leqslant 1\}$.

Example

Suppose that the stochastic process $X_t = A + Bt + Ct^2, t \in (-\infty, +\infty)$, where the random variables A, B and C are independent, and each has a normal distribution with mean 0 and variance σ^2. Show that X_t is a Gaussian process.

Gaussian Processes

Example

Suppose that the process $\{X_t\}_{t\in T}$ is defined by

$$X_t = U\cos\omega_0 t + V\sin\omega_0 t, \quad t \geqslant 0,$$

where ω_0 is a constant, U and V are two independent r.v.'s with

$$E(U) = E(V) = 0, \quad E(U^2) = E(V^2) = \sigma^2.$$

Show that $\{X_t\}_{t\in T}$ is a Gaussian process and determine the first-order and second-order probability density functions.

Contents

1. Independent-Increment Processes
2. Poisson Process
3. Gaussian Processes
4. Brownian Motion and Wiener Processes

Brownian Motion and Wiener Processes

Brownian motion is a sophisticated random number generator, based on a process in plants discovered by Robert Brown in 1827. He found that small particles suspended in a fluid were in continuous movement and thus, described it as Brownian motion. His discovery did not receive much attention for a long time, until before the turn of the 20th century when Leon Gouy's conviction and research (that Brownian motion constituted a clear demonstration of the existence of molecules in continuous motion) brought it to the attention of the physics world. However, all research in nineteenth-century remained at the qualitative level.

Brownian Motion and Wiener Processes

It was only in 1905 when a quantitative analysis was brought about, where Einstein succeeded in stating the mathematical laws governing the movements of particles on the basis of the principles of the kinetic-molecular theory of heat. According to this theory, bodies of microscopically visible size suspended in a liquid will perform irregular thermal movements called Brownian molecular motion, which can be easily observed in a microscope. Brownian motion was then more generally accepted because it could now be treated as a practical mathematical model. As a result, many scientific theories and applications related to it have been developed and they subsequently play major roles in the world of physics.

Brownian Motion and Wiener Processes

Definition

A real-valued process $\{W_t\}_{t \geqslant 0}$ *is said to be a Wiener process (also called Brownian motion) with parameter* σ^2*, if*
(i) $W_0 = 0$,
(ii) *the increments*

$$W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \cdots, W_{t_m} - W_{t_{m-1}}$$

are independent for all $0 = t_0 < t_1 < \cdots < t_m$,
(iii) $W_t - W_s$ *has a normal distribution with mean 0 and variance* $\sigma^2(t-s)$ *for all* s, t, $0 \leqslant s < t$.

Brownian Motion and Wiener Processes

From (ii) and (iii), we know the process $\{W_t\}_{t \geqslant 0}$ has stationary, independent increments. A Wiener process with initial value $W_0 = x$ is gotten by adding x to a standard Wiener process. From (i) and (iii) we know

$$W_t \sim N(0, \sigma^2 t).$$

Property

Suppose that W_t is a Wiener process. Then its mean and auto-correlation function are given by

$$\mu_W(t) = 0, \quad C_W(s,t) = R_W(s,t) = \sigma^2 \min(s,t), \quad s,t \geqslant 0.$$

Brownian Motion and Wiener Processes

By using the property (iii) in Definition, we have

Property

The Wiener process W_t is a Gaussian process.

Proof. For all $0 \leqslant t_1 < t_2 < \cdots < t_n$, $a_1, a_2 \ldots, a_n \in \mathbb{R}$, and for all integers $n > 0$, we have the linear combination

$$\sum_{k=1}^{n} a_k W_{t_k} = \sum_{k=1}^{n} b_k (W_{t_k} - W_{t_{k-1}}), \quad t_0 = 0,$$

where $b_k = \sum_{i=k}^{n} a_i, k = 1, 2, \ldots, n$. Since W_t has independent increments, we know $\sum_{k=1}^{n} a_k W_{t_k}$ has the normal distribution. Thus we obtain that W_t is a Gaussian process. $\square$

Brownian Motion and Wiener Processes

In general, a stochastic process with stationary, independent increments is called a Lévy process. The Wiener process is the intersection of Gaussian process and Lévy process.

Example

Suppose that $\{W_t\}_{t\geqslant 0}$ *is a Wiener process. Find the autocorrelation functions of the stochastic processes* $Y_t = W_{t+l} - W_t,\ t \geqslant 0$, *(* $l > 0$ *is a constant).*

Brownian Motion and Wiener Processes

Property

Suppose that W_t is a Wiener process. Then the following processes are all Gaussian processes.

(i) $U_t = -W_t$ *for* $t \geqslant 0$.

(ii) $Y_t = \begin{cases} tW_{1/t} & \text{for } t > 0, \\ 0 & \text{for } t = 0. \end{cases}$

(iii) $X_t = \dfrac{1}{c} W_{c^2 t}$ for $c > 0$.

Brownian Motion and Wiener Processes

One of the many reasons that Brownian motion is important in probability theory is that it is, in a certain sense, a limit of rescaled simple random walks.

Why is this important?

First, it explains, at least in part, why the Wiener process arises so commonly in nature. Many stochastic processes behave, at least for long stretches of time, like random walks with small but frequent jumps. The argument above suggests that such processes will look, at least approximately, and on the appropriate time scale, like Brownian motion.

Second, it suggests that many important "statistics" of the random walk will have limiting distributions, and that the limiting distributions will be the distributions of the corresponding statistics of Brownian motion.

Summary

- Suppose that $\{X_t\}_{t\geqslant 0}$ is an independent-increment process and $X_0 = 0$. Then the covariance function

$$C_X(s,t) = Var_X[\min(s,t)].$$

- Let N_t be a Poisson process with rate λ. Then
(i) $\mu_N(t) = \lambda t$,
(ii) $C_N(s,t) = Var_N[\min(s,t)] = \lambda \min(s,t)$,
(iii) $R_N(s,t) = \lambda \min(s,t) + \lambda^2 st$,
where

$$\min(s,t) = \frac{s+t-|s-t|}{2}.$$

Summary

- A Gaussian process is completely determined by its mean and covariance functions.
- $\{X_t\}_{t\in T}$ is a Gaussian process if and only if any linear combination of any finite number of random variables has a normal distribution, i.e., for distinct $t_1, t_2, \cdots, t_n \in T$ and $c_1, c_2, \cdots, c_n \in \mathbb{R}$, the random variable $c_1 X_{t_1} + c_2 X_{t_2} + \cdots + c_n X_{t_n}$ has a (univariate) normal distribution.
- Suppose that W_t is a Gaussian process. Then its mean and autocorrelation function are given by

$$\mu_W(t) = 0, \quad C_W(s,t) = R_W(s,t) = \sigma^2 \min(s,t), \quad s,t \geqslant 0.$$

The end

Thank you !